

Storage Tanks Design and Construction

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Pressure vessels dept. scope of work

Scope of work

◎ **Proposal phase.**

Clients send a request to EPC for Projects execution in their plants

Request includes:

- Preliminary study documents
- FEED (Front End Engineering Design) documents

As an EPC company, we do a more detailed study and budget estimate
For the project

After project award, contract is signed and Engineering phase starts



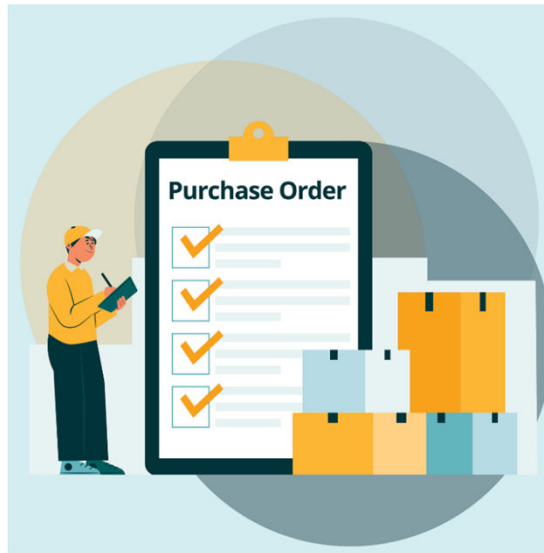
Scope of work

- ④ **Quotation phase**
MDSs and MRQ are being prepared during this phase and sent to bidders
- ④ **Clarification Phase**
After Receiving offers from bidders vague point are covered through clarification cycle
- ④ **Technical Evaluation Phase**
After clarification phase we decide which bidder satisfy our requirements from technical point of view
- ④ **Commercial evaluation documents**
Any point agreed during clarification cycle is added to MDSs and MRQ for commercial evaluation

EPC scope of work

- ④ **Issue Purchase documents**

MDSs and MRP are issued for the successful supplier



Why Storage Tanks

Why Storage Tanks

- ⦿ **Used to store**
 - Crude oil
 - Intermediate and refined products
 - Gas
 - Chemicals and water
- ⦿ **Tanks are built in different types and sizes and of various construction materials depending on their intended use.**



Storage Tanks classification

Storage Tanks classification

◎ Tanks Are Classified According To:

- Design Code
- Service Conditions
- Type
- Dimensions



Storage Tanks classification

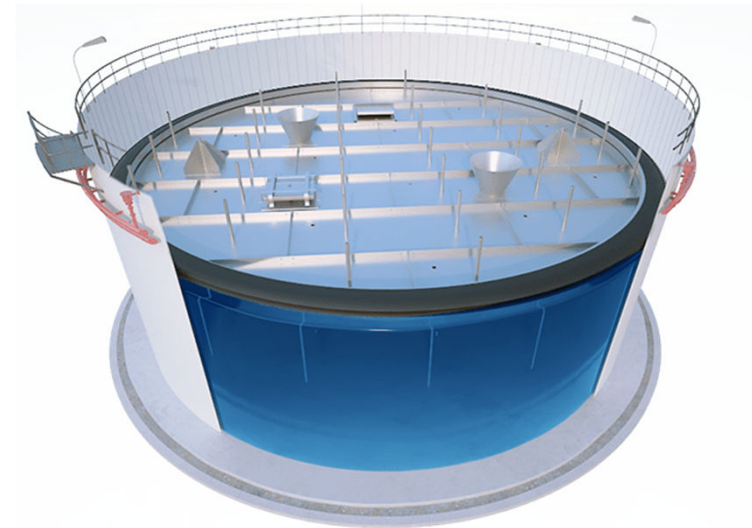
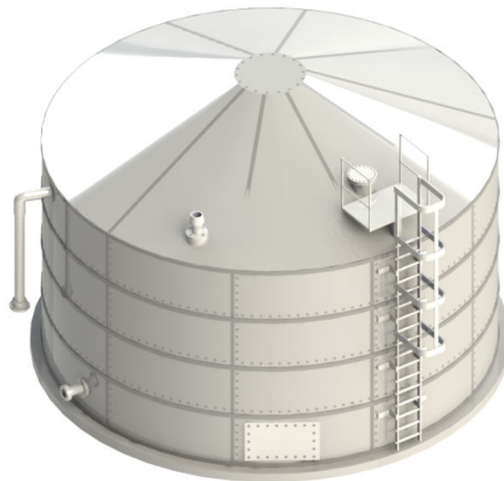
◎ Tanks Are Classified According To:

- Type:
- Metallic tanks
- FRP tanks



Storage Tanks classification

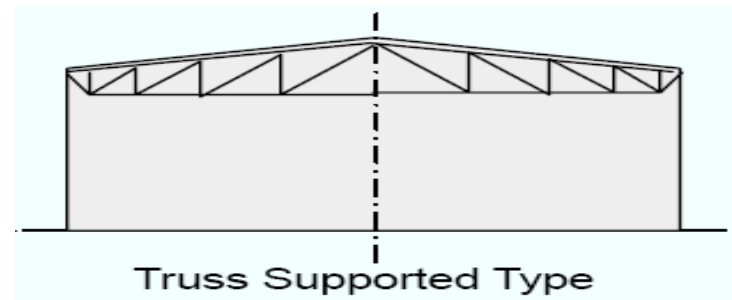
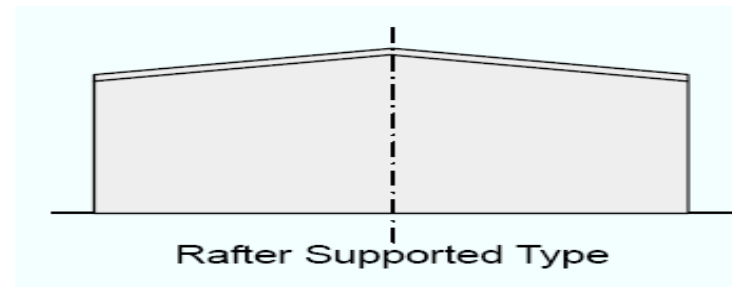
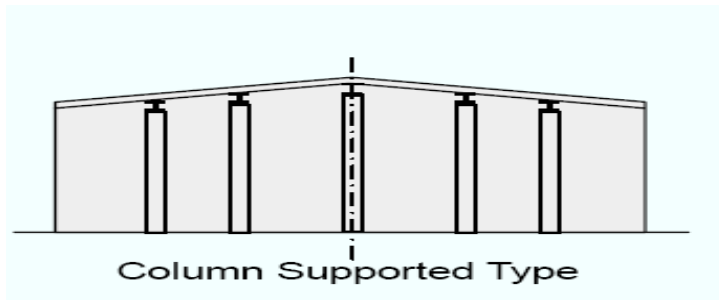
- ◎ **Design Code:**
 - Atmospheric Tanks (API 650)
 - Design Pressure < 2.5 PSI G.
 - Fixed Cone Roof
 - Floating Roof



Storage Tanks classification

◎ Design Code:

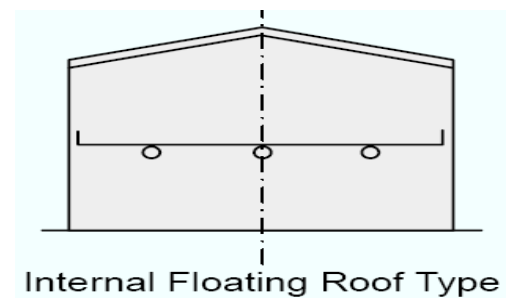
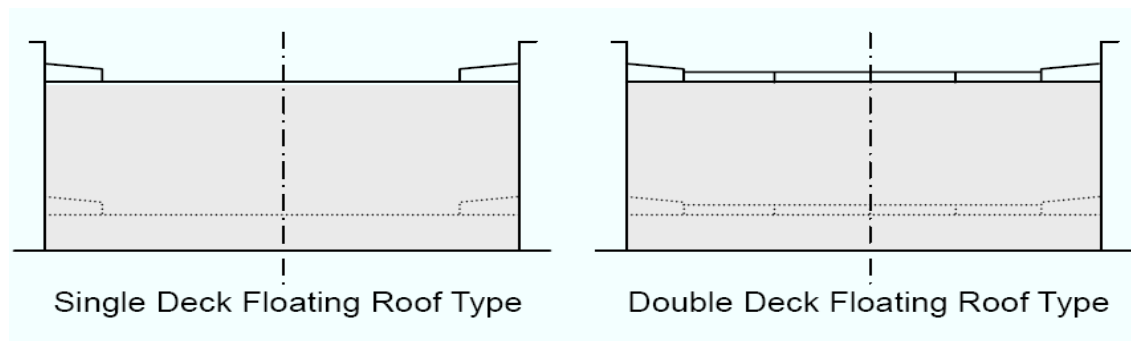
- Atmospheric Tanks (API 650)
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Storage Tanks classification

◎ Design Code:

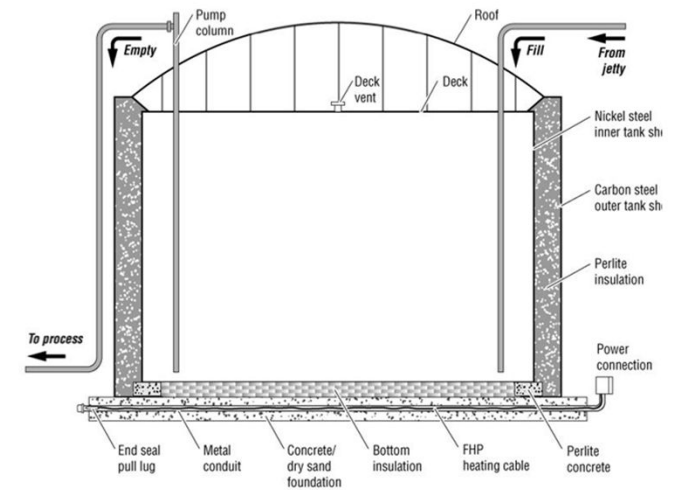
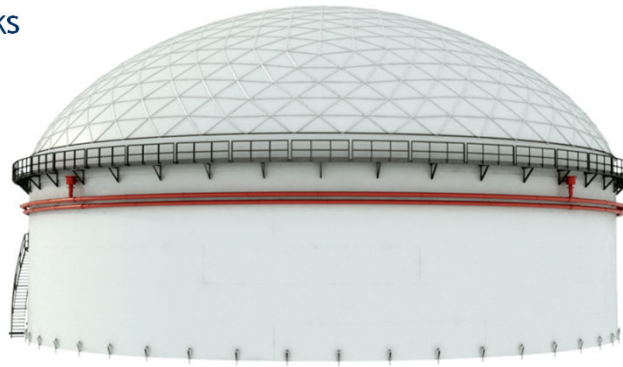
- Atmospheric Tanks (API 650)
- Design Pressure < 2.5 PSI G.
- Fixed Cone Roof
- Fixed Dome roof
- Floating Roof



Storage Tanks classification

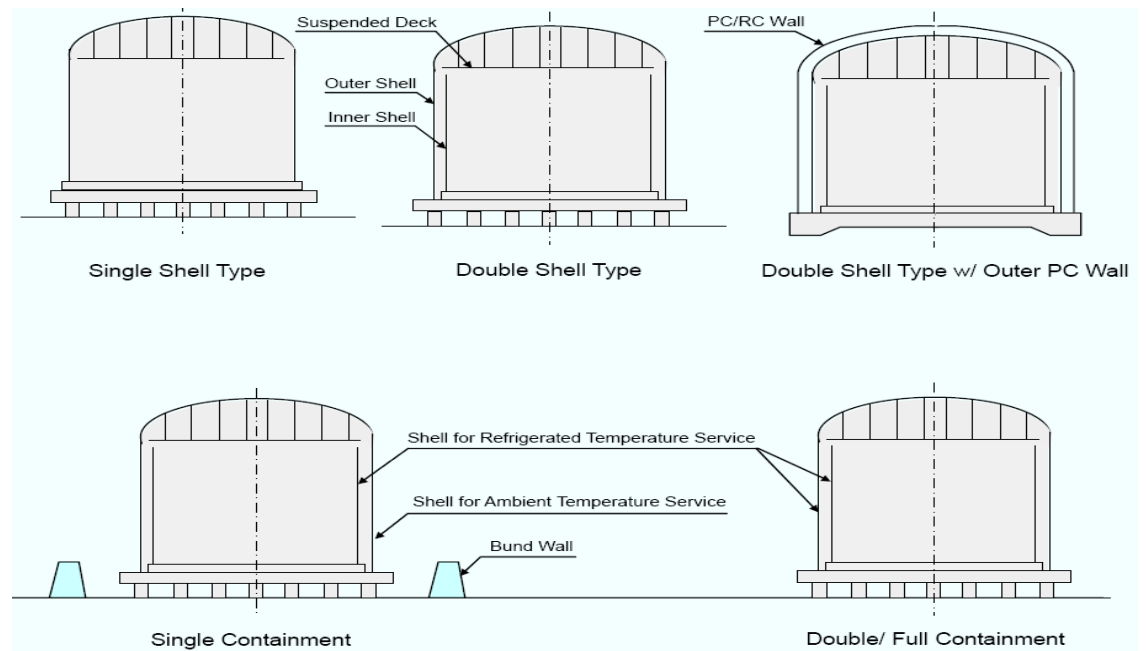
◎ Design Code:

- Low Pressure Tanks (API 620)
Design Pressure < 15 PSI G.
Dome
Cryogenic Low Temp Tanks
(ex. double wall tanks)



Storage Tanks classification

- ◎ **Design Code:**
 - Low Pressure Tanks (API 620)
 - Design Pressure < 15 PSI G.
 - Dome
 - Cryogenic Low Temp. Tanks



Storage Tanks classification

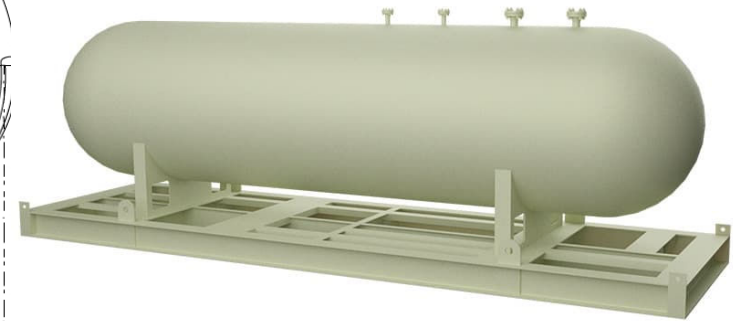
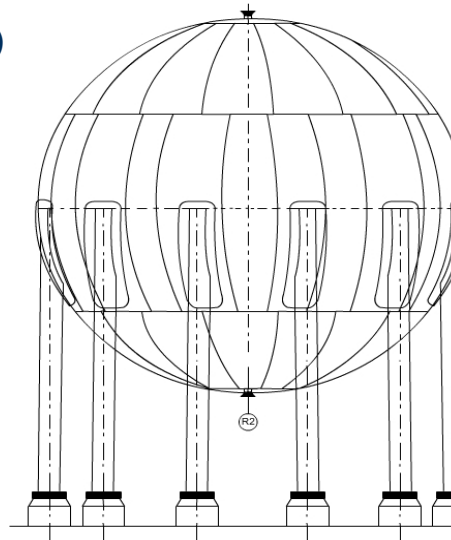
◎ Design Code:

- High Pressure Tanks (ASME Code)

 - Design Pressure > 15 PSI G.(103 kPa)

 - Spherical Tanks

 - Bullet Tanks



Storage Tanks classification

- ◎ **Type:**
 - Shop Fabricated (Less than 6m diameter.)
 - Field Erected

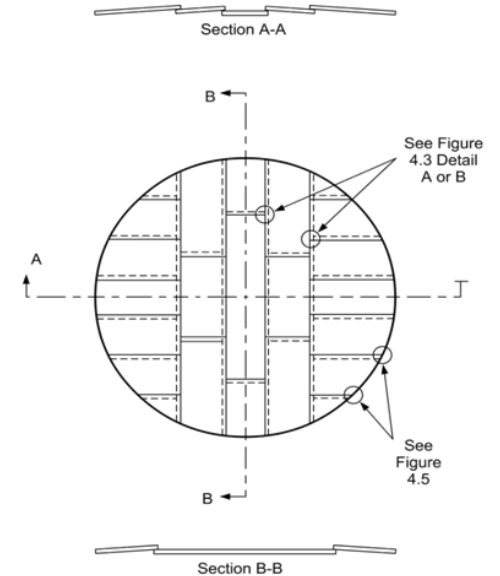
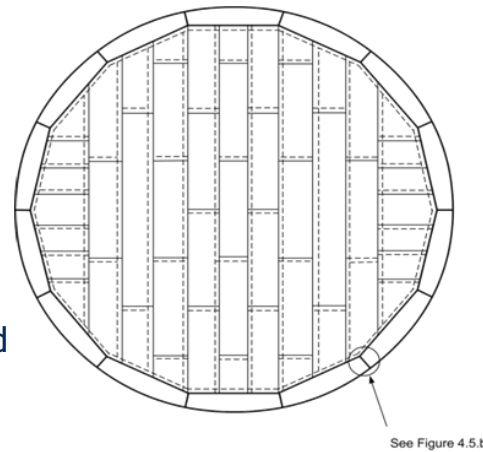


Storage Tanks Main Elements

Storage Tanks Main Elements

⦿ Bottom Plates:

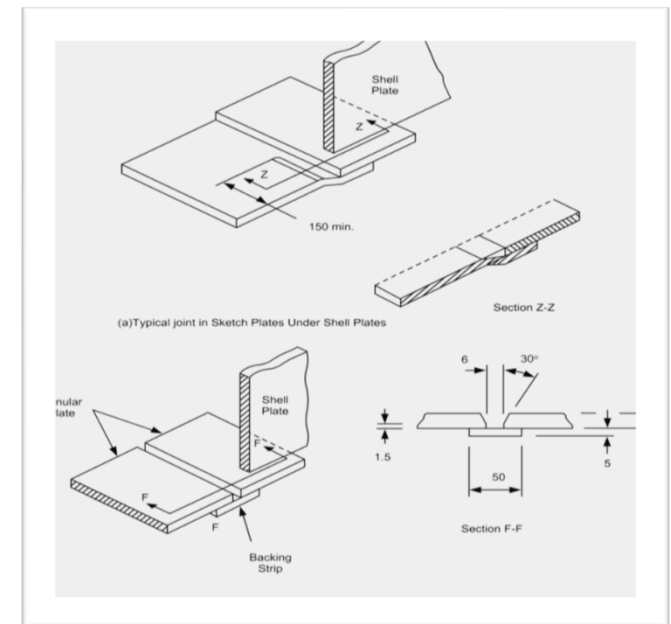
- Made of several rectangular plates, surrounded by a set of shaped plates
- The plates slightly overlap each other and are pressed locally at the corners, where three plates meet.
- Lapped and fillet welded joints are preferred to butt welded joints.



Storage Tanks Main Elements

⦿ Bottom Plates:

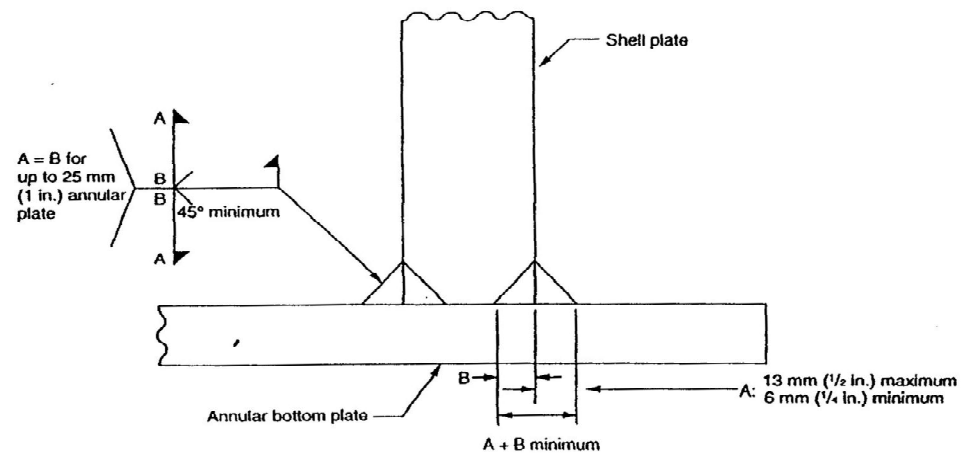
- For larger tanks (over 12.5 m diameter) a ring of annular plates is provided around the group of rectangular plates.
- The radial joints between the annular plates are butt welded, rather than lapped, because of the ring stiffening which the plates provide to the bottom of the shell.



Storage Tanks Main Elements

◎ Shell-to-Bottom Fillet Welds

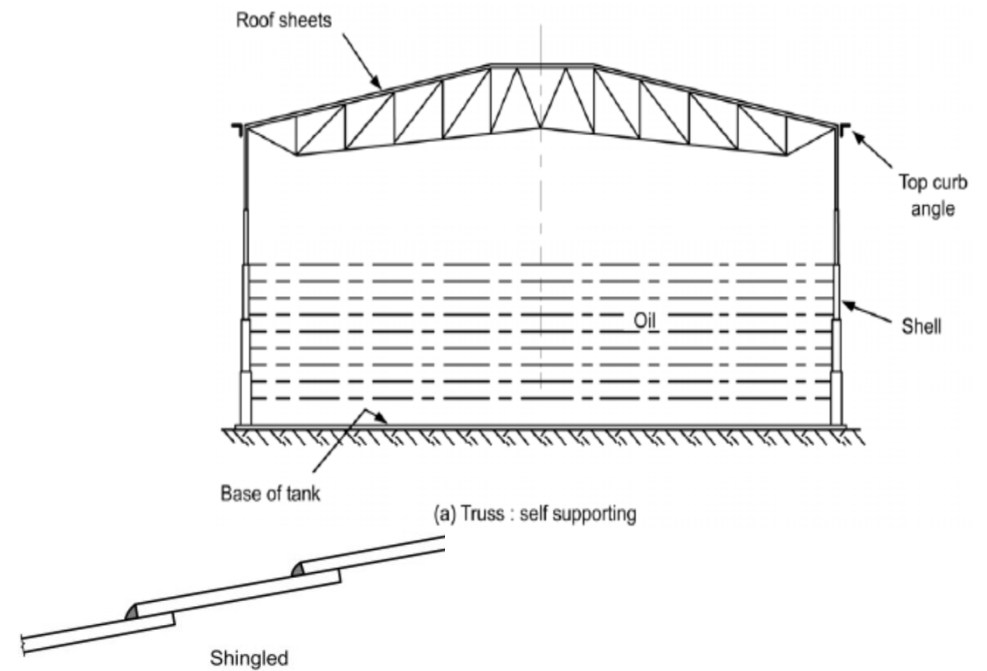
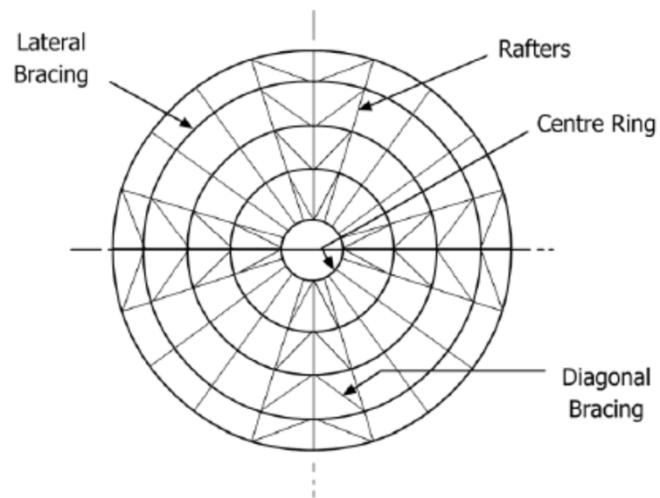
-For tanks over 12.5 m diameter or when otherwise required by code e.g. seismic loading, a ring of annular plates is provided around the group of rectangular plates.



Storage Tanks Main Elements

⦿ Roof Plates

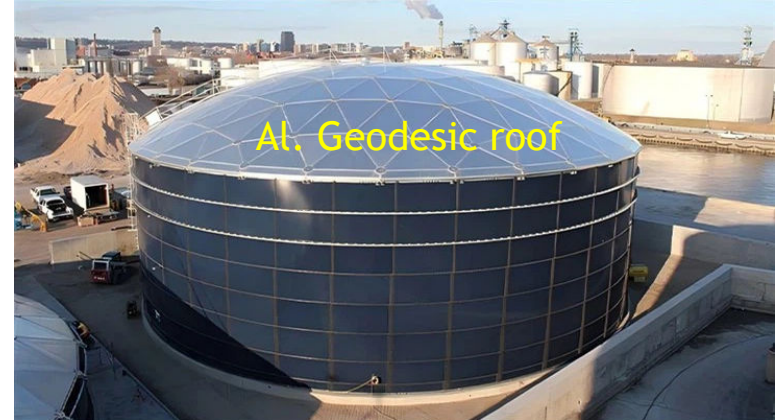
-Tanks roofs can either be in conical, or dome or umbrella layout



Storage Tanks Main Elements

⦿ Roof Plates

-Tanks roofs can either be conical, dome, geodesic or umbrella layout

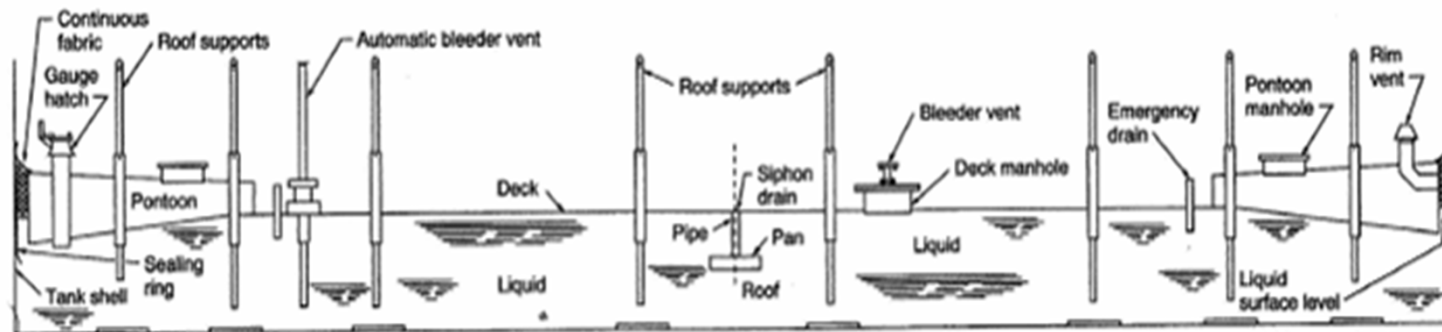


Floating roof Tanks

External Floating roof tanks

External Floating roof

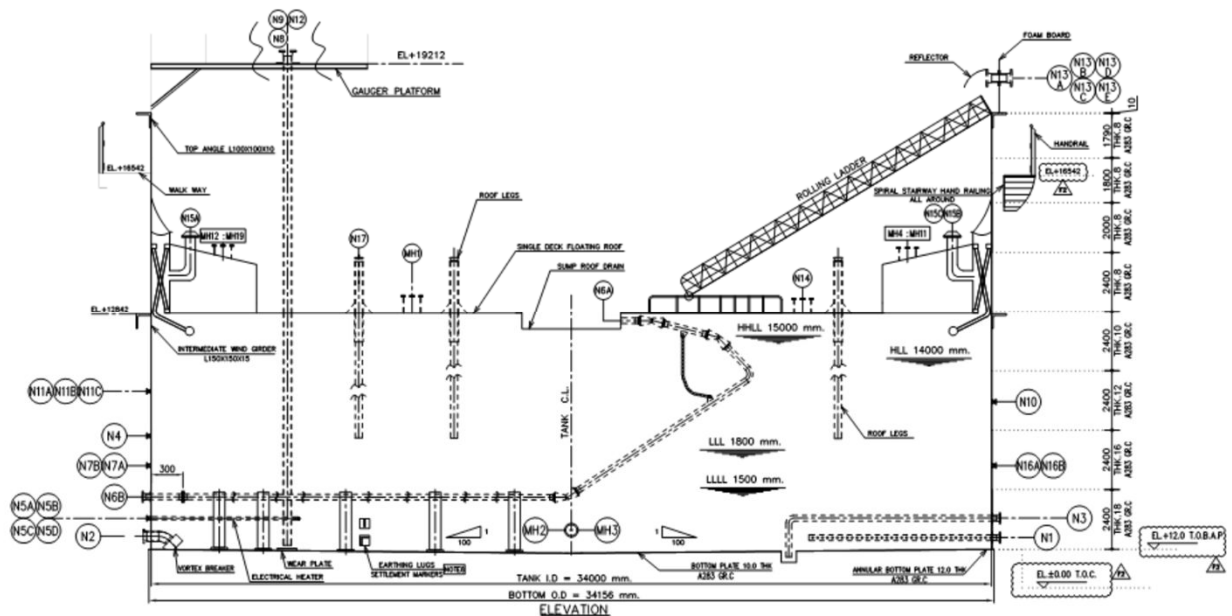
- Floats on the surface of the oil, rather than being carried above the oil on structural steel supports.
- It rides up and down inside the tank shell as the oil level changes.



External Floating roof tanks

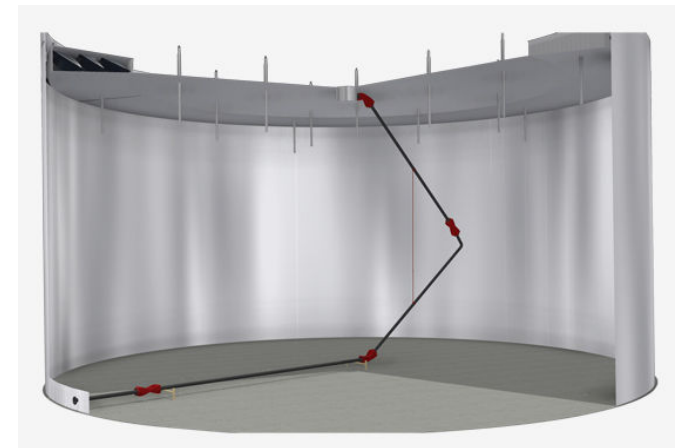
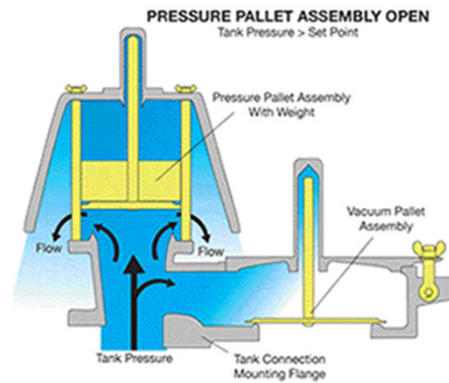
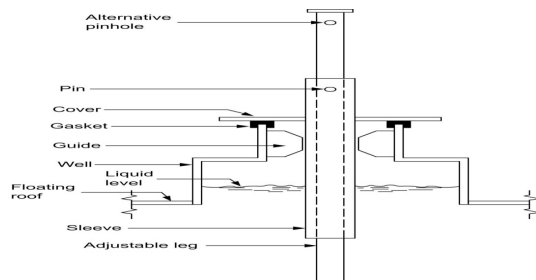
● Floating roof

- Floats on the surface of the oil, rather than being carried above the oil on structural steel supports.
- It rides up and down inside the tank shell as the oil level changes.



Floating roof Tanks Accessories

- ◎ Appurtenances are auxiliary equipment items that are added to storage tanks to perform specific functions such as:
 - Vents
 - Roof Drains
 - Vacuum breaker



External Floating roof Tanks Accessories

- ⦿ **Auxiliary equipment items that are added to floating tanks such as:**
 - Gauger platform - Rolling ladder
 - Guide pole(e.g. for Guided Wave Rader)

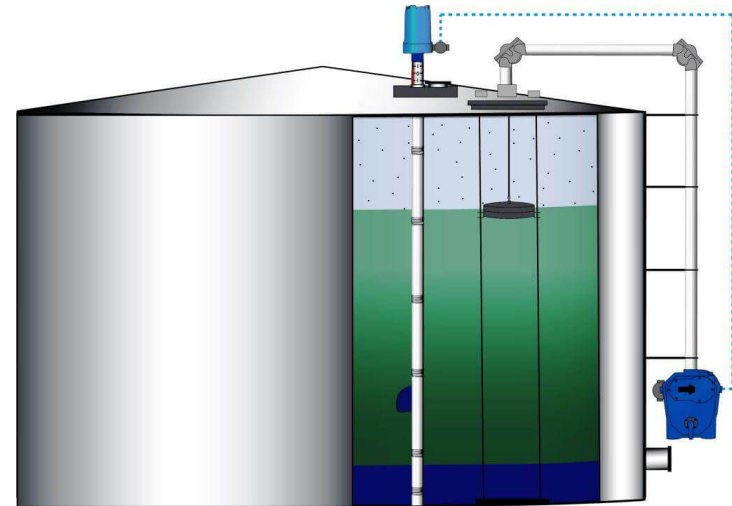


Storage Tanks Accessories

Storage Tanks Accessories

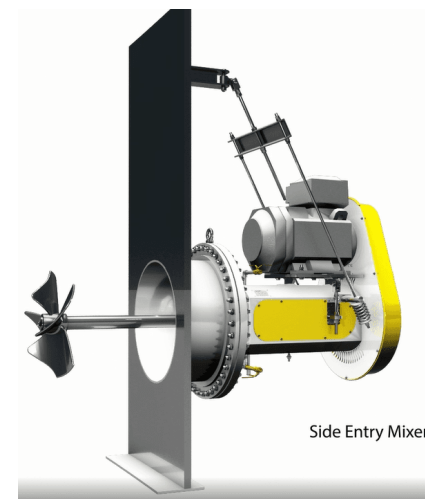
- ◎ Appurtenances are auxiliary equipment items that are added to storage tanks to perform specific functions such as:

- Platforms, Ladders, and Stairways
- Mechanical level gauge
- Vent with goose neck
- Gauge hatch



Storage Tanks Accessories

- ◎ Appurtenances are auxiliary equipment items that are added to storage tanks to perform specific functions such as:
 - Mixers
 - Manways
 - Cleanout doors



Site erection

Site erection

- ◎ **Bottom plate erection**
 - prefabricated sections shall arrive at site
 - sections: rolled as required and ready for welding.



Site erection

◎ Tank erection

- Foundation works
- Bottom plate welding
- Prefabricated sections receiveal at site. Sections: rolled as required and ready for site welding.



Site erection

◎ Shell erection methods

- Jacking

- Conventional method

*note consider wind during erection



Site erection

⦿ Roof erection methods

- Roof crane Lifting
- Roof Air lifting



Thank You
Any Questions?